

This Zenodo deposit contains a publicly available description of the Dataset:

"Quantitative DIA-based proteomic analysis of lysosomes (Lyso-IP) from VPS35[D620N] Mouse Lung with LRRK2 inhibition".

Dataset Description:

Proteomic analysis of lung lysosomes from VPS35[D620N] mice treated in vivo with MLI-2 (100 mg/kg, 2 h). Data acquired on an Orbitrap Exploris 480 and processed using DIA-NN 1.8.1.

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Dataset Name: alessi-mouse-ms-p-lung-vps35-d620n-dmso-mli2, v0.1

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ASAP Project: Mapping the LRRK2 Signalling Pathway and Its Interplay with Other Parkinson's Disease Components.

Project Description: LRRK2 targets and modifies a set of enzymes known as Rab GTPases, triggering new biological events by creating new protein: protein interactions. We aim to decipher what controls the activity of LRRK2 and to explore, in precise molecular detail, how this enzyme affects three major cellular structures (cilia, lysosomes and mitochondria) implicated in Parkinson's disease. We showed that mutant LRRK2 triggers a series of molecular changes that cause new sets of proteins to interact. Our goal is to use a combination of state-of-the-art approaches to understand the consequences of these new interactions on the biology of three important subcellular compartments: primary cilia, lysosomes, and mitochondria.

Submission Date: 2026-04-30

This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the [User Manual](#).

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This Zenodo deposit was created by the ASAP CRN Cloud staff on behalf of the dataset authors. It provides a citable reference for a CRN Cloud Dataset